

**AMENDMENT NO. 1 MARCH 2003
TO
IS 3597 : 1998 CONCRETE PIPES — METHODS OF
TEST**

(Second Revision)

(Page 4, clause 8) — Substitute the following for the existing:

‘8 PERMEABILITY TEST

8.1 Prestressed Concrete Pipes and Precast Concrete Pipes

This test shall be done on outside surface of the pipe. No additional treatment of any type shall be done on the pipe before permeability test is carried out. For Prestressed Concrete Pipe, the test shall be conducted at 3 places on coating and for Precast Concrete Pipe at 2 places simultaneously, immediately after curing is completed (see Fig. 4). In case this is done later, the pipe shall be kept wet for 48 hours prior to test. For plain/flush ended precast pipes, it shall be carried out about 300 mm away from both ends.

8.1.1 Procedure

The dry surface of the pipe shall be scrapped by wire brush and loose particles, if any, removed. Sealant shall then be applied to the lower portion of the cup and cup shall be pressed on the pipe. After hardening of sealant, water shall be filled in the cup with wash bottle. The glass tube with rubber cork shall then be fixed in the cup as shown in Fig. 4. Water in the tube shall then be filled using wash bottle and air shall be allowed to escape during filling. Precaution shall be taken, so that water does not leak either from cup ends or from the rubber stopper.

8.1.2 Initial Absorption

Water shall be filled up to zero mark and reading shall be taken at every half hour interval up to two hours. The drop in water level in the stand pipe at the end of two hours in the initial absorption.

Amend No. 1 to IS 3597 : 1998

8.1.3 Final Permeability

Fill the water in the stand pipe again up to zero mark and take the reading at one hour interval up to 4 h. The absorption in the fourth hour, that is, difference between fourth and third hour readings is the final permeability. The average of tests conducted at three places for prestressed concrete pipe and two places for precast concrete pipe shall be expressed in cm^3 as final permeability.

Criteria for acceptance is the final permeability.'

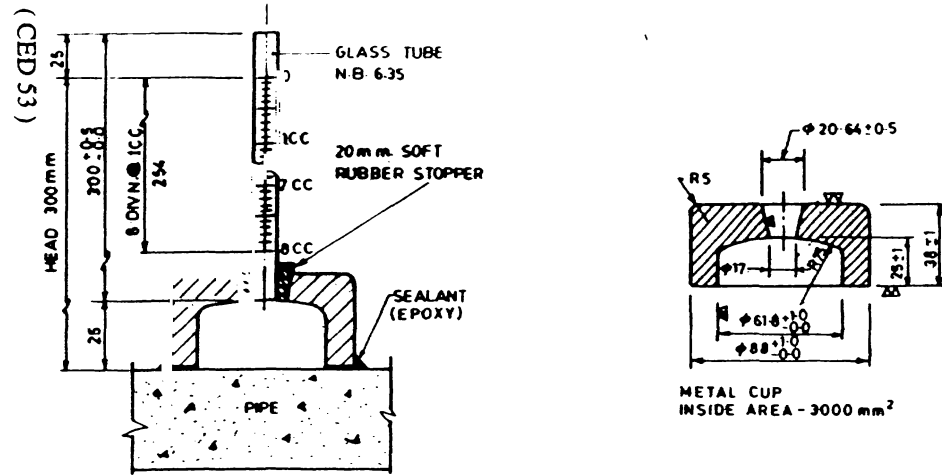
(Page 4, clause 7.2.4) — Substitute the following for the existing:

'7.2.4 Pressure shall be maintained for 1 min + 30 s for each 10 mm of wall thickness (for precast concrete pipes wall thickness shall be full barrel wall thickness, whereas it shall be core thickness, in case of prestressed concrete pipe) or for twice that entire period if the application of pressure resulted in the formation of beads of water on the pipe surface.'

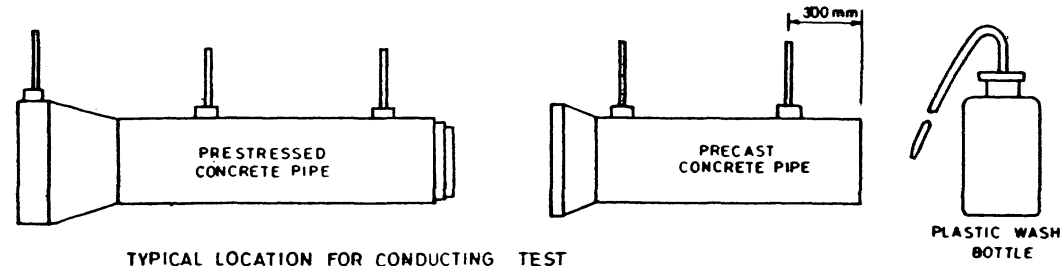
(Page 4, clause 7.2.6, line 4) — Insert the following at the end of first sentence:

'(for precast concrete pipes wall thickness shall be full barrel wall thickness, whereas it shall be core thickness, in case of prestressed concrete pipe)'

(Page 5, Fig. 4) — Substitute the figure given on page 3 for the existing:



3



TYPICAL LOCATION FOR CONDUCTING TEST

All dimensions in millimetres.

FIG. 4 METHOD FOR PERMEABILITY TESTING OF CONCRETE PIPES

Amend No. 1 to IS 3597:1998

Printed at Dee Kay Printers, New Delhi